Spring 25 - C291/E111 - Systems Programming with C & Unix Assignment 1

Q.1 Hello World (10 marks)

Write a simple program that prints the sentence:

Welcome to "C291"
Systems Programming with C & Unix

Make sure to use the correct escape sequence to put the cursor onto the next line at the end of these sentences.

Note: The solution for this problem should be named as solution_1.c and it should be under C291-Spring-25 /assignment1 where assignment1 would be the new directory you would create.

Q.2 Printing a Custom Sentence (20 marks)

Write a program that will take input from the user to print a custom sentence. You will need to create 3 separate variables and ask the user to enter values for each. These integer variables will represent the month, year, and your age. With these, print a sentence of the following format:

It is month m of the year y, and you are a year old!(where m, y and a are the month, year and age entered by the user).

Example 1:
Enter current month: 3
Enter current year: 2020
Enter your age: 19
Output: It is month 3 of the year 2020 and you are 19 years old!

Example 2:
Enter current month: 8
Enter current year: 2023
Enter your age: 100
Output: It is month 8 of the year 2023 and you are 100 years old!

Note: The solution for this problem should be named as solution_2.c and it should be under C291-Spring-25 /assignment1 where assignment1 would be the new directory you would create.

Q.3 Simple Arithmetic (30 marks)

Write a program that performs simple mathematical operations given 2 integers by the user.

Create 2 integer variables ‘a’ and ‘b’ and ask the user to input values for these. Next, you will use these variables to print the result of the following operations:

- a + b (Addition)
- a – b (Subtraction)
- a * b (Multiplication)
- a % b (Remainder)

Print these answers on separate lines in sentences.

Example 1:
Input a: 25
Input b: 5
Output:

The sum of 25 and 5 is 30.
The difference of 25 and 5 is 20.
The product of 25 and 5 is 125.
The remainder of 25 and 5 is 0.

Example 2:
Input a: 8
Input b: 10
Output:

The sum of 8 and 10 is 18.
The difference of 8 and 10 is -2.
The product of 8 and 10 is 80.
The remainder of 8 and 10 is 8.

Note: The solution for this problem should be named as solution_3.c and it should be under C291-Spring-25 /assignment1 where assignment1 would be the new directory you would create.

Q.4 Circle Area, Diameter and Circumference (20 marks)

Compute the diameter, circumference, and area of a circle for which you accept the radius 'r' from the user. Use the value 3 for π. Use the following formulas (r is the radius): diameter = 2r, circumference = 2πr and area = πr^2 Perform each of these calculations inside the printf statement(s).

You will print these answers on separate lines in sentences.

Example 1:
Input: Enter the radius: 2

Output:

The Area of circle with radius 2 is 12.
The Diameter of circle with radius 2 is 4.
The Circumference of circle with radius 2 is 12.

Note: The solution for this problem should be named as solution_4.c and it should be under C291-Spring-25 /assignment1 where assignment1 would be the new directory you would create.

Q.5 Find the nth term of the Arithmetic Progression (20 marks)

Write a C program to find the nth term of an AP series. You will take 3 inputs from user, 'a' will be the starting number, 'd' will be the common difference and 'n' will be the nth term you need to find.

The formula to find the nth term of an AP series is

$$AP = a + (n - 1) * d$$

Example 1:

Input:

Enter the starting number of AP series: 2
Enter the common difference: 1
Enter the nth term to be found: 5

Output:

The 5th term of the series is: 6

Note: The solution for this problem should be named as solution_5.c and it should be under C291-Spring-25 /assignment1 where assignment1 would be the new directory you would create.